WASP-43b

R-1621

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_210306` · created 2026-08-02T04:37:23+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459279.7976 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.021
Transit depth (Rp/Rs)^2	3.79 +/- 0.84 [%]
Orbital Inclination (inc)	79.44 +/- 1.2
Airmass coefficient 1 (a1)	0.9001 +/- 0.0095
Airmass coefficient 2 (a2)	0.077 +/- 0.035
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	#2 - [372, 235]
Optimal Aperture	3.93
Optimal Annulus	6.49
Transit Duration (day)	0.0431 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.021 vs 0.1594 (deviation 1.7σ)
Duration fit vs published	0.0431 d vs 0.0512 d (deviation 1.25σ)
Mid-time O–C	-0.0 min
Depth SNR	9.3
Residual scatter	1.05 %

Light curve

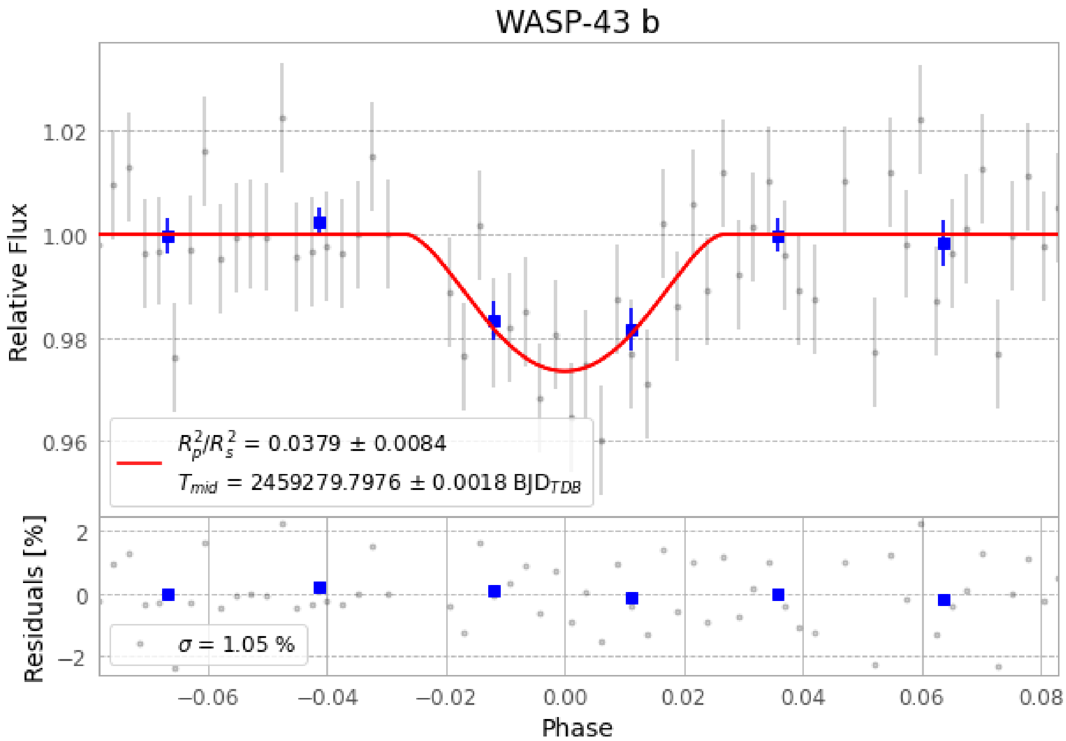

Figure 1 — FinalLightCurve_WASP-43 b_2021-03-05.png (SHA-256 8914fc4a68b9b5a1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [505, 192], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 54fcb7b937db248e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

64 FITS frames (42 MB), `WASP-43210306052712.FITS` → `WASP-43210306083612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210306.log (67206d1a8d99…), FinalLightCurve_WASP-43 b_2021-03-05.png (8914fc4a68b9…), FinalLightCurve_WASP-43 b_2021-03-05.csv (48aa013bdc3d…)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-02 04:37Z.